2026s-CS758-Lecture05

Topics

- Successive minima
- Shortest-vector problem (SVP) and its friends
- Lagrange's algorithm for 2D-SVP

Sources

1. Intro to Lattice Algorithms and Cryptography, Lecture 2
2. Intro to Lattice Algorithms and Cryptography, Lecture 3
3. Intro to Lattice Algorithms and Cryptography, Lecture 4
4. The LLL Algorithm: Survey and Algorithms, Chapter 02

Plan

- Plan/Recall
  - In the last lecture,
    - We defined the minimum distance $\lambda_1(\Lambda)$ of a lattice Λ and saw how it is relevant to cryptography and coding theory
    - Then we went on to establish upper and lower bounds for the minimum distance. In particular, we saw Minkowski's first theorem, which upper bounds λ_1 in terms of the determinant (which is basis-independent):

$$\lambda_1(\Lambda(\bar{B})) \leq \sqrt{n} \cdot det(\bar{B})^{1/n}$$

      - Essentially a consequence of pigeonhole principle via Blichtfeldt's theorem and Minkowski's convex-body theorem
  - In today's lecture:
    - We first define *successive minima*, which generalises the notion of minimum distance
    - Then we will introduce computational problems related to minimum distance: the shortest vector problem (SVP) and its approximate variant
    - In general SVP is a hard problem, we'll see that for the 2D case, there is an efficient algorithm: the Lagrange's algorithm. Along the way, we will learn reduced bases
- Successive Minima
  - Recall the definition of λ_1: it can be seen as the radius of the smallest ball that contains a non-zero lattice point
  - Informally, λ_i is the radius of the smallest ball that contains i *linearly-independent* lattice points

- **Example 1.** Consider $\Lambda := \Lambda((1,0)^t, (3,4)^T)$. Note that $\lambda_1(\Lambda) = 1$ since $(1,0)^t \in \Lambda$. Note that $(2,0)^t$ is also in Λ, but it is linearly dependent on $(1,0)^t$. It seems that $\lambda_2(\Lambda) = 4$ since $(0,4) \in \Lambda$.
- More formally:
  - **Definition 1.** For $1 \leq i \leq m$, we define

    $$\lambda_i(\Lambda) := \inf_{r \geq 0} dim(\Lambda \cap B_r^n) \geq i,$$

    where B_r^n is the n-dimensional ball of radius r
    - Note that $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_m$
  - **HW1.**
    - Let $\Lambda \subset \mathbb{R}^n$ be a m-rank lattice. Show that there exists vectors $\bar{y}_1, \cdots, \bar{y}_m \in \Lambda$ such that $\|\bar{y}_i\| = \lambda_i$.
    - Does Λ have a basis $\bar{b}_1 \cdots, \bar{b}_m$ such that $\|\bar{b}_i\| = \lambda_i$? This is true for Λ in Example 1, but is it true in general?
- Minkowski also proved the following theorem bounding the *geometric mean* of successive minima:
  - **Theorem 1 (Minkowski's second theorem, see [1], Theorem 21).** For any full-rank lattice Λ of dimension n,

    $$(\Pi_{i=1}^n \lambda_i(\Lambda))^{1/n} \leq \sqrt{n} \cdot det(\Lambda)^{1/n}$$

    - Note that Theorem 2 is a stronger bound than Minkowski's first theorem since the mean is at least λ_1.

- Shortest-Vector Problem and its Friends
  - We are now ready to define some computational problems on lattices
    - **Definition 2 (SVP: Shortest Vector Problem, [3], Definition 1).**
      - *Input.* A basis $\bar{B}$ of lattice Λ
      - *Solution.* A vector $\bar{y} \in \Lambda$ such that $\|\bar{y}\| = \lambda_1(\Lambda)$
    - **Definition 3 (SVP$_\alpha$: Approximate SVP, [3], Definition 1).**
      - *Input.* A basis $\bar{B}$ of a lattice Λ of dimension n and an approximation factor $\alpha(\cdot)$
      - *Solution.* A vector $y \in \Lambda$ such that $\|\bar{y}\| \leq \alpha(n)\lambda_1(\Lambda)$
    - **Definition 4 (GapSVP$_\gamma$: gap SVP, [4], Definition 1).**
      - *Input.* A basis $\bar{B}$ of lattice Λ, a distance $d \in \mathbb{Q}$ and an gap factor $\gamma(n)$
      - *Decision.* Is $\lambda_1(\Lambda) > \gamma(n)d$ or $\lambda_1(\Lambda) \leq d$?
    - **Definition 5 (SIVP$_\alpha$: Approximate Shortest Independent Vector Problem, [4], Definition 2).**
      - *Input.* A basis $\bar{B}$ of lattice Λ and an approximation factor $\alpha(\cdot)$
      - *Solution.* m linearly-independent lattice vectors $\bar{y}_1, \cdots, \bar{y}_m$ such that $\|\bar{y}_i\| \leq \alpha(n)\lambda_n(\Lambda)$
  - What do we know about these problems? (Source)

- SVP
  - Ajtai showed that SVP is NP-hard using a decision-to-search (randomised) reduction from *any* NP problem to SVP
    - Priyanshu will cover basics of complexity theory in Recitation 2.
  - We don't know how to base cryptography on SVP
- SVP$_\alpha$
  - Khot showed that SVP is NP-hard for any constant approximation factor, i.e, $\alpha := O(1)$
  - For approximation factors $\alpha(n) := \sqrt{n/\log(n)}$ and $\alpha(n) := O(\sqrt{n})$, SVP$_\alpha$ was shown to be in $\mathbf{coNP}$ and $\mathbf{coAM}$ respectively by Aharonov and Regev and Goldreich and Goldwasser
  - For approximation factors $\alpha(n) := n$, it seems we can do cryptography (I was wrong!).
  - In the next lecture, we will see the LLL algorithm, which solves SVP$_\alpha$ for $\alpha := 2^{O(n)}$.
- Here is a very nice schematic by Vinod Vaikuntanathan capturing all of the above:

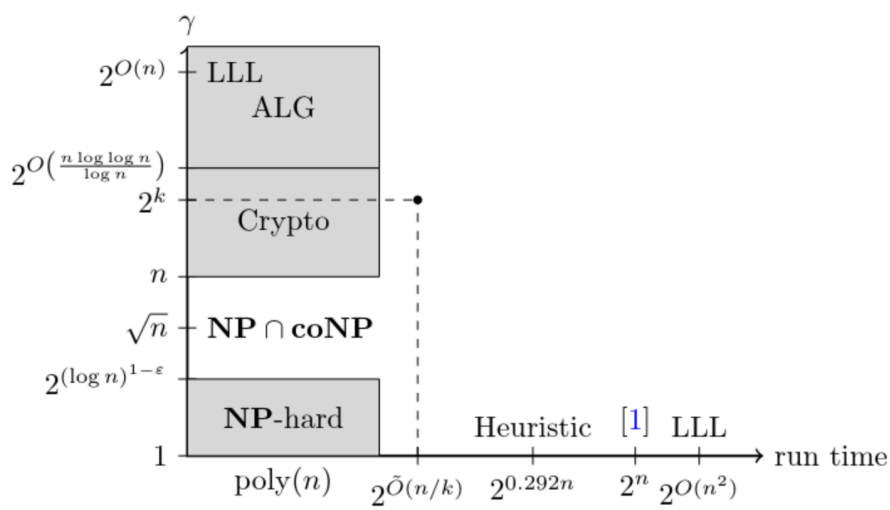

Figure 1: The complexity landscape of SVP$_\gamma$.

- GapSVP$_\gamma$
  - We can build cryptography from GapSVP$_\gamma$ for $\gamma(n) := O(n)$: see the work of Micciancio and Regev
- SIVP$_\alpha$
  - Ajtai showed how to base cryptography on approximate SIVP for $\alpha := poly(n)$. We will see this result in Module II

- Lagrange's algorithm for 2D-SVP
  - In this section we will see that although (exact) SVP is $\mathbf{NP}$-hard, the hardness only really kicks in in higher dimensions. That is, we will see a polynomial-time algorithm for SVP when $n = 2$, called Lagrange's algorithm. In this process we will

introduce the notion of a "reduced" basis. The techniques used will be useful in the next lecture when we discuss the celebrated LLL algorithm for *approximate* SVP.

- *Reduced basis.* Informally a basis is "reduced" if consists of short, almost orthogonal vectors. There are several notions of reduced. In this lecture, we will consider "Lagrange" reduced basis for $\mathbb{R}^2$, while in the next lecture we will consider "LLL-reduced" basis for $\mathbb{R}^n$.
  - **Definition (Wristwatch Condition).** A basis $(\bar{b}_1, \bar{b}_2)$ for a lattice $\Lambda \subset \mathbb{R}^2$ is Lagrange-reduced iff

    $$\|\bar{b}_1\| \leq \|\bar{b}_2\|$$

    and

    $$|\langle \bar{b}_1, \bar{b}_2 \rangle| \leq \|\bar{b}_1\|^2/2.$$

    - Geometrically, it looks like a wrist-watch: see figure.
  - In the following claim, we show that a Lagrange-reduced basis is in fact the "shortest basis"!
  - **Claim 1.** For a Lagrange-reduced basis $(\bar{b}_1, \bar{b}_2)$, $\|\bar{b}_1\| = \lambda_1(\Lambda)$ and $\|\bar{b}_2\| = \lambda_2(\Lambda)$.
  - *Proof.*
    1. First we show that $\|\bar{y}\| \geq \|\bar{b}_1\|$ for any non-zero $\bar{y} \in \Lambda$. Since $\bar{y} \in \Lambda$, there exists $x_1, x_2 \in \mathbb{Z}$ such that $\bar{y} = x_1\bar{b}_1 + x_2\bar{b}_2$. Now we have,

       $$\begin{aligned} \|\bar{y}\|^2 &= \|x_1\bar{b}_1 + x_2\bar{b}_2\|^2 \\ &= x_1^2\|\bar{b}_1\|^2 + 2x_1x_2\langle \bar{b}_1, \bar{b}_2 \rangle + x_2^2\|\bar{b}_2^2\| \\ &\geq x_1^2\|\bar{b}_1^2\| - x_1x_2\|\bar{b}_1\|^2 + \|\bar{b}_2^2\| \\ &\geq (x_1^2 - x_1x_2 + x_2^2)\|\bar{b}_1\|^2 \\ &\geq \|\bar{b}_1\|^2, \end{aligned}$$

       where
       - The first inequality holds by the second clause of wristwatch condition, i.e., $\langle \bar{b}_1, \bar{b}_2 \rangle \geq -\|\bar{b}_1\|^2/2$,
       - The second inequality holds by its first clause, and
       - The last inequality holds since $\bar{y}$ is non-zero and both $x_1 = 0$ and $x_2 = 0$ cannot hold
    2. Next, we show that $\|\bar{y}\| \geq \|\bar{b}_2\|$ whenever $x_2 \neq 0$ (i.e., $\bar{y}$ is not linearly dependent on $\bar{b}_1$.
       - **HW2.** Prove the second part of Claim 1.
- *The algorithm.*
  - The algorithm is similar to Euclid's GCD algorithm, which we now recall:
    - **Algorithm 1**:
      - *Input*: $a, b \in \mathbb{Z}$
      - *Output*: $gcd(a, b)$
      - Algorithm:

1. If $|a| \leq |b|$ then swap a and b
2. Else, while $b \neq 0$ do:
    1. Compute $r := a - \lfloor a/b \rceil b$, where $\lfloor \cdot \rceil$ is the function that rounds to the nearest integer (say breaking ties upwards)
    2. Set $a := b$ and $b := r$
3. Output a

- **Example 2.** Running Algorithm 1 on $(56, 315)$, results in the sequence $(315, 56)$, $(56, 35)$, $(35, 21)$, $(21, 14)$, $(14, 7)$, $(7, 0)$

- The idea is to carry out Gram-Schmidt *with rounding* and swap the resulting basis vectors as long as $\bar{b}_1$'s length is greater than $\bar{b}_2$
    - **Algorithm 2**:
        - *Input*: Basis $(\bar{b}_1, \bar{b}_2) \in \mathbb{R}^{2\times 2}$
        - *Output*: Lagrange-reduced basis $(\bar{b}_1, \bar{b}_2) \in \mathbb{R}^{2\times 2}$
        - Algorithm:
            1. If $\|\bar{b}_1\| < \|\bar{b}_2\|$ then swap $\bar{b}_1$ and $\bar{b}_2$
            2. Else, repeat
                1. Compute $\bar{r} := \bar{b}_1 - \lfloor \mu_{1,2} \rceil \bar{b}_2$, where if you recall $\mu_{1,2} := \langle \bar{b}_1, \bar{b}_2 \rangle / \langle \bar{b}_2, \bar{b}_2 \rangle$
                2. Set $\bar{b}_1 := \bar{b}_2$ and $\bar{b}_2 := \bar{r}$.
                Until $\|\bar{b}_1\| \leq \|\bar{b}_2\|$.
            3. Output $(\bar{b}_1, \bar{b}_2)$
    - **Example 3.** On running Algorithm 2 on $((1,5)^t, (6,21)^t)$ results in the sequence $((6,21)^t, (1,5)^t)$, $((1,5)^t, (2,1)^t)$ and $(2,1)^t, (-1,4)^t)$. See figure
- **Theorem 2.** Algorithm 2 runs for $O(\log(\|\bar{b}_1\| / \sqrt{det(\Lambda)})$ iterations and outputs a Lagrange-reduced basis.
- *Proof.*
    - **Claim 2.** Algorithm 2 outputs a Lagrange-reduced basis.
    - *Proof.*
        - **Pop-Quiz 1.** Why does the output forms a basis of Λ? Because we are only doing the column operations corresponding to unimodular matrix
        - **Pop-Quiz 2.** Why does the wristwatch condition hold?
            - First, note that Algorithm 1 *terminates* since $\|\bar{b}_1\|$ diminishes in each iteration: we are replacing $\bar{b}_1$ with the shorter of the basis vectors, i.e., $\bar{b}_2$, and $\bar{b}_2$ is shrinking since we are replacing it with "rounded" orthogonal projections of $\bar{b}_1$ (similar to Gram-Schmidt). Since termination can only happen when $\|\bar{b}_1\| \leq \|\bar{b}_2\|$, the first wristwatch condition holds.

- To see why, let's denote the basis at the end of each iteration by $(\bar{b}_1', \bar{b}_2')$ observe by construction that

$$\begin{aligned}\langle \bar{b}_1', \bar{b}_2' \rangle &= \langle \bar{b}_2, \bar{b}_1 - \lfloor \mu_{1,2} \rceil \bar{b}_2 \rangle \\ &= \langle \bar{b}_1, \bar{b}_2 \rangle - \lfloor \mu_{1,2} \rceil \|\bar{b}_2\|^2\end{aligned}$$

  Showing that the above expression is at most $\|\bar{b}_1'\|^2/2 = \|\bar{b}_2\|^2/2$ is the same as showing

$$\langle \bar{b}_1, \bar{b}_2 \rangle / \|\bar{b}_2\|^2 - \lfloor \mu_{1,2} \rceil \leq 1/2.$$

  However, this is true since $\mu_{1,2} := \langle \bar{b}_1, \bar{b}_2 \rangle / \|\bar{b}_2\|^2$ and definition of rounding.

- **Claim 3.** Algorithm 2 runs for $O(\log(\|\bar{b}_1\| / \sqrt{det(\Lambda)})$ iterations.
  - **HW3.** Show that the length of $\bar{b}_1$ reduces by a constant factor in each iteration. What exactly is the constant